



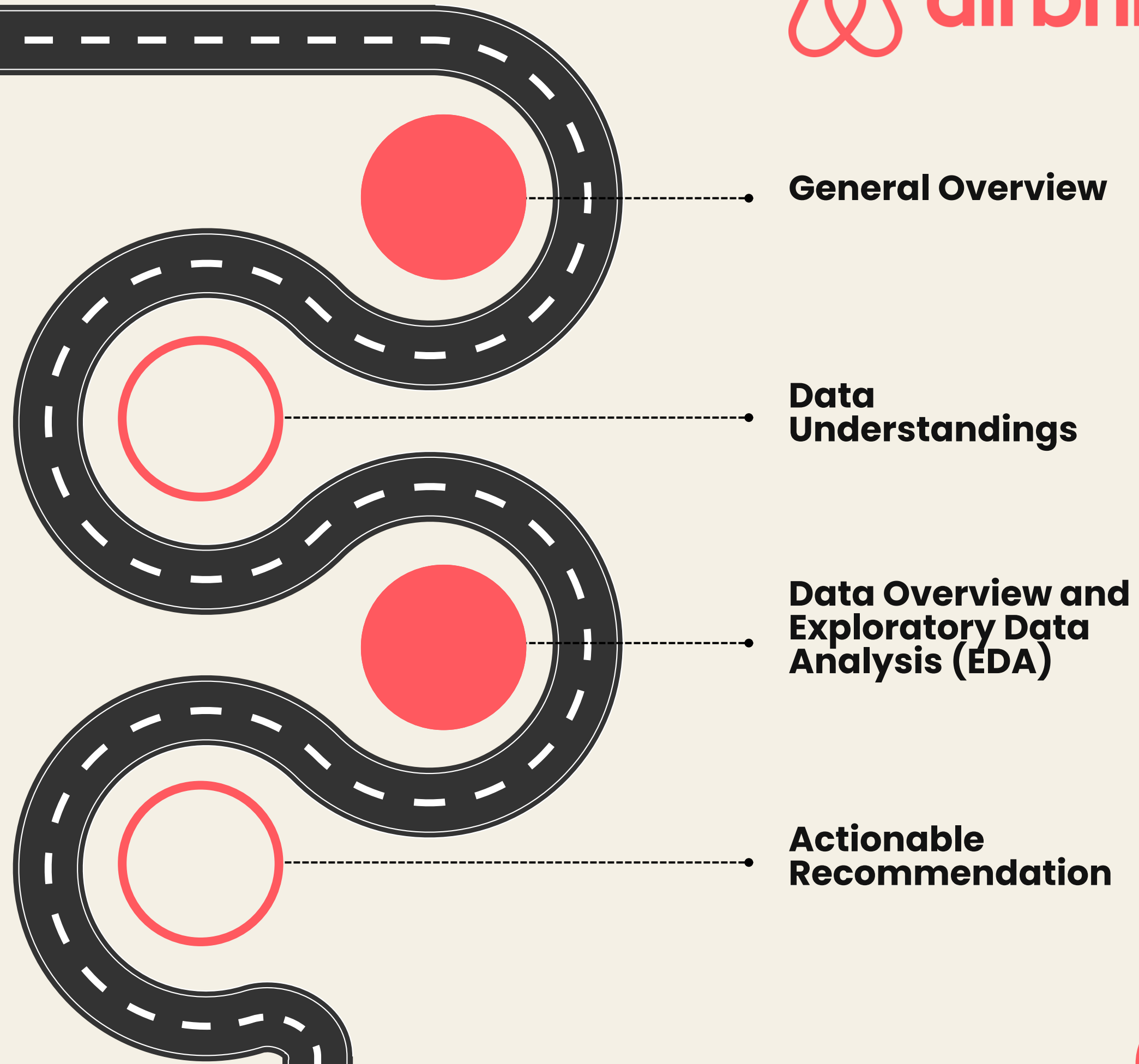
Airbnb Bangkok Listing Insights Report

2012 – 2022 :

Data-Driven Insights to Improve Occupancy, Pricing, and Revenue Optimization

Medina Alifia Juniarto

Agenda



General Overview

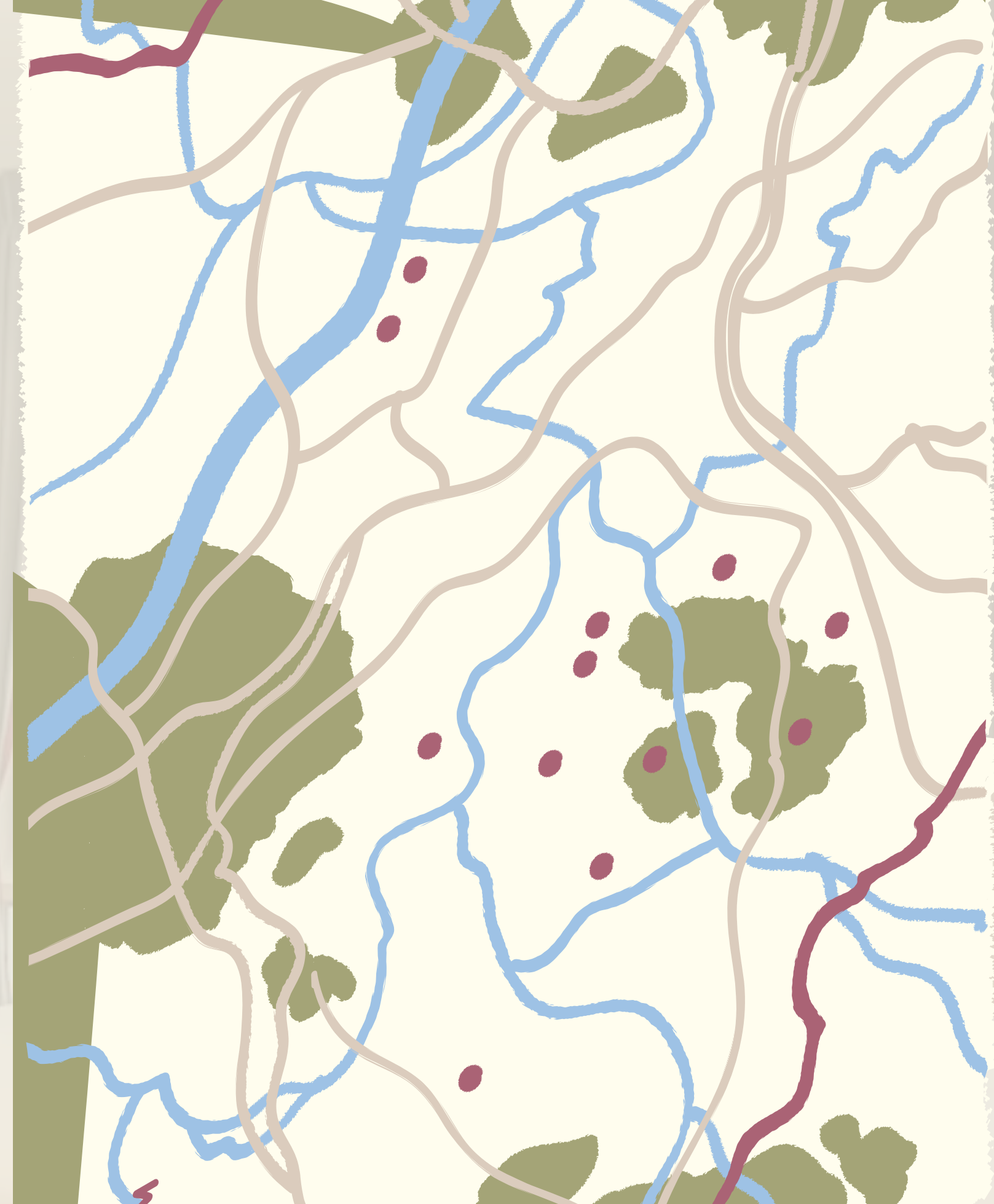
Problem Statement

Background: Uneven Performance

Airbnb Bangkok, the top tourism hub, from 2012 to 2022, shows a steady growth, a sharp COVID-19 downturn in 2020, and a rebound by 2022. Despite recovery, Airbnb listings show inconsistent performance. Prices often do not reflect demand, many listings sit empty despite high availability, and guest reviews are low for a large portion of properties. Multi-listing hosts also struggle with uneven results across their portfolios.

Problem Statement

How can Airbnb hosts in Bangkok leverage data-driven strategies to **optimize pricing, improve occupancy rates, increase guest engagement, and achieve consistent performance** across multiple listings in a post-pandemic market?



General Overview

Company Profile

Airbnb is....

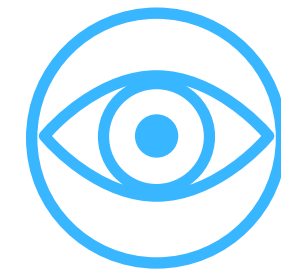
A U.S. based platform that connects **hosts** offering spaces with **guests** looking for short – or long-term stays and travel experiences. It earns revenue through **service fees on each booking and operates** in over 190 countries.

Bangkok, one of the most visited cities globally, is a key market for Airbnb's growth in Southeast Asia.



Mission

To create a world where anyone can belong anywhere.

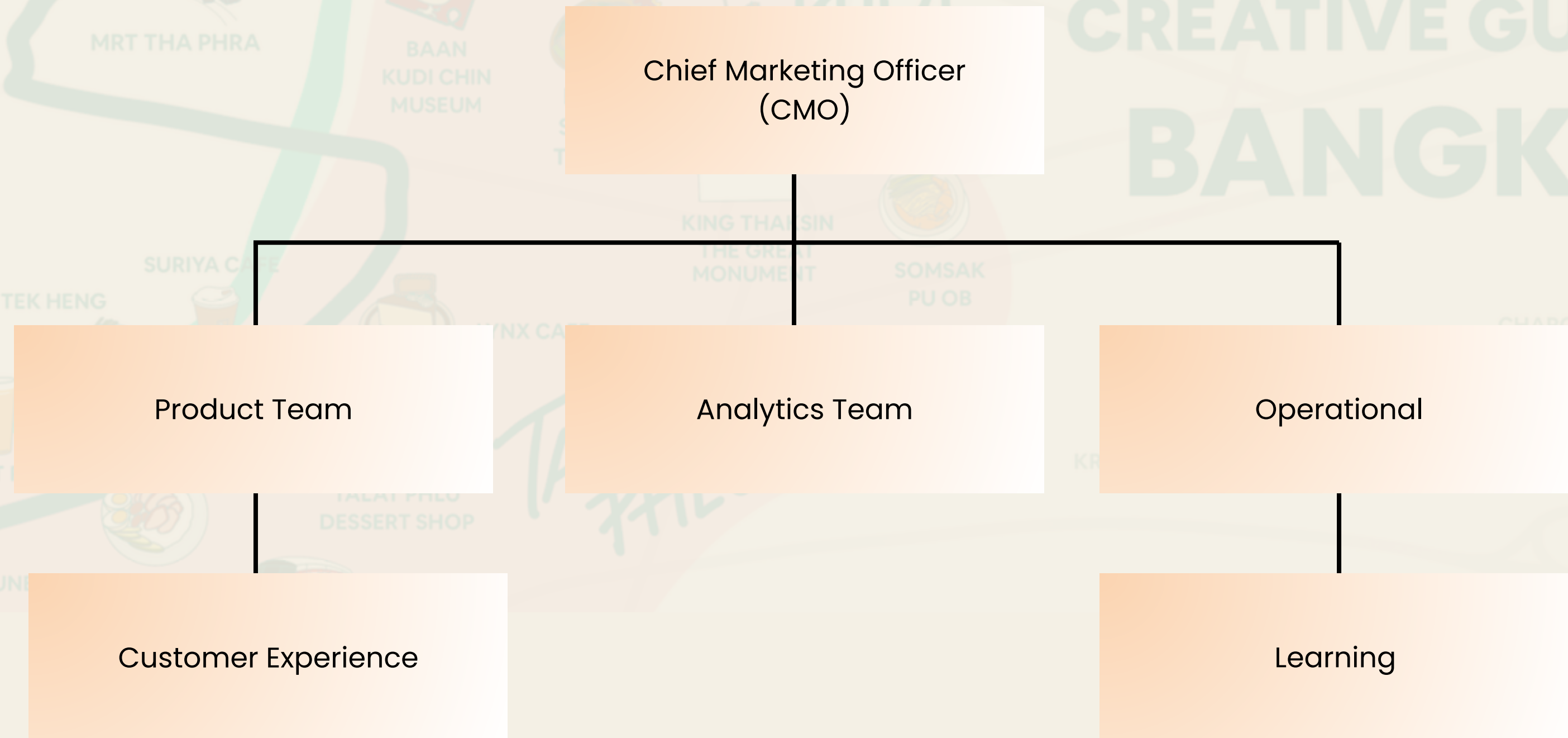


Vision

Belong Anywhere.

General Overview

Stakeholders



Data Understanding

Data Classification



ID Listing Identification

- **id** : Unique identifier for each Airbnb listing.
- **name** : Title or name of the listing.



Host Information

- **host_id** : Unique identifier for the host.
- **host_name** : Name of the host (usually first name).
- **calculated_host_listings_count** : Total number of listings the host manages in the area.



Property Specifications

- **room_type** : Entire home, Private room, Shared room.
- **price** : Total price in local currency.
- **minimum_nights** : Minimum number of nights required to book.
- **price_per_night** : Price per night



Availability

- **availability_365** : Number of available days in the next year.



Performance Metrics

- **number_of_reviews** : Total number of reviews received.
- **number_of_reviews_ltm** : Number of reviews received in the last 12 months.
- **last_review** : Date of the most recent review.
- **estimated_revenue** : The price x number of reviews
- **last_review(Year)** : The most recent year reviews.



Location Details

- **neighbourhood** : Geocoded neighbourhood based on official shapefiles.
- **latitude** : Latitude coordinate (WGS84).
- **longitude** : Longitude coordinate (WGS84).

Airbnb Bangkok Data Overview



Scale & Market Overview



Total Hosts
6,659



Total Listings
15,854

Listings per Room Type:

- Entire Home/Apt : 8,912
- Private Room : 5,770
- Hotel Room : 649
- Shared Room : 523



Top 5 Revenue Drivers by Neighbourhood



★★★★★

Khlong Toei

฿ 193,431,184

Huai Khwang

฿ 154,609,413

Vadhana

฿ 105,941,999

Bang Rak

฿ 43,748,616

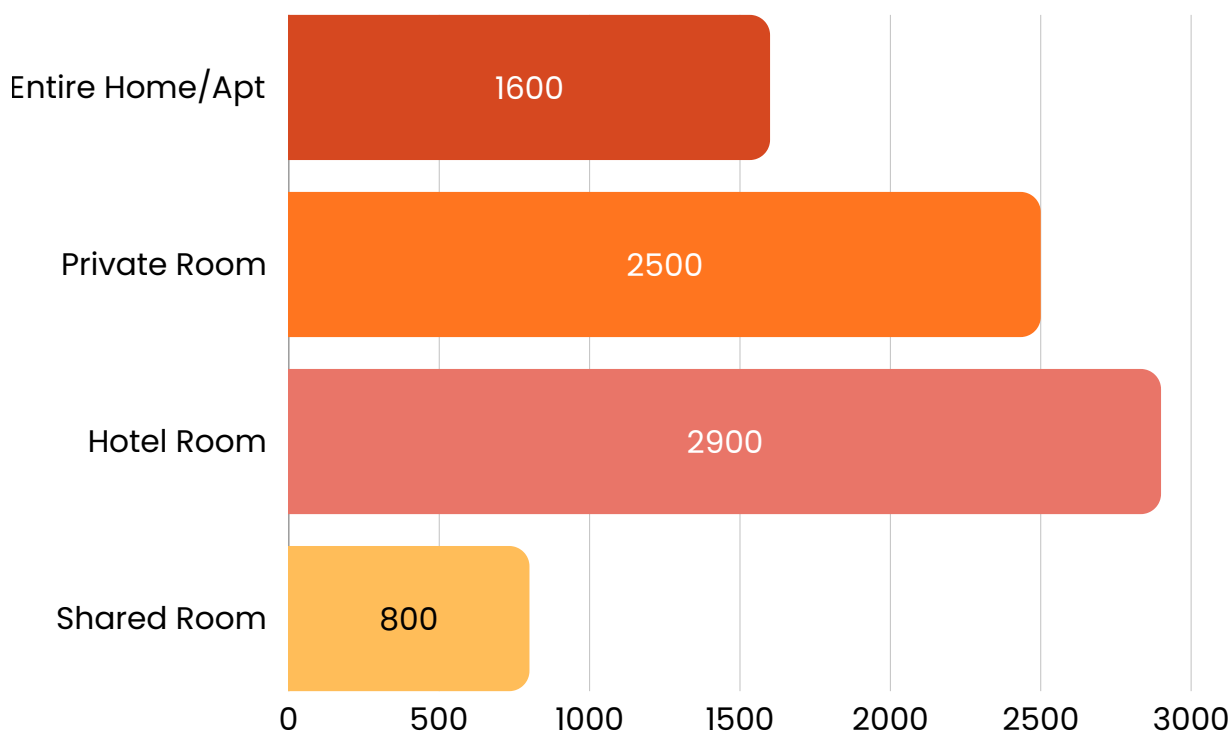
Ratchathewi

฿ 42,375,064



Pricing Landscape

Average Price Per Night According Room Type (฿)



Occupancy Rate & Reviews

Avg. Occupancy Rate
33.05%

Monthly Review Rate
51.62%

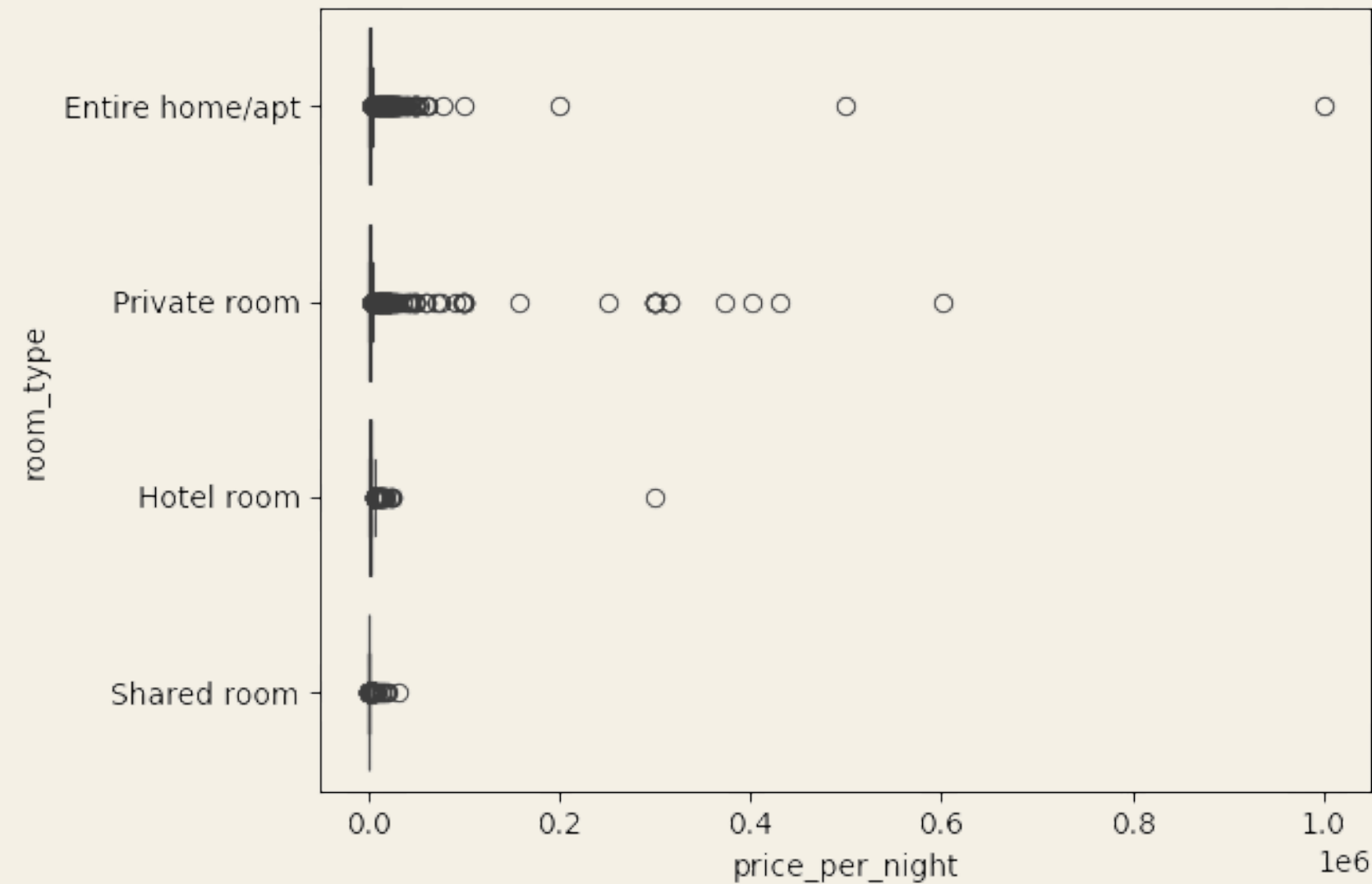
Hosts with <1 review/month
92.13%

EXPLORATORY DATA ANALYSIS (EDA)

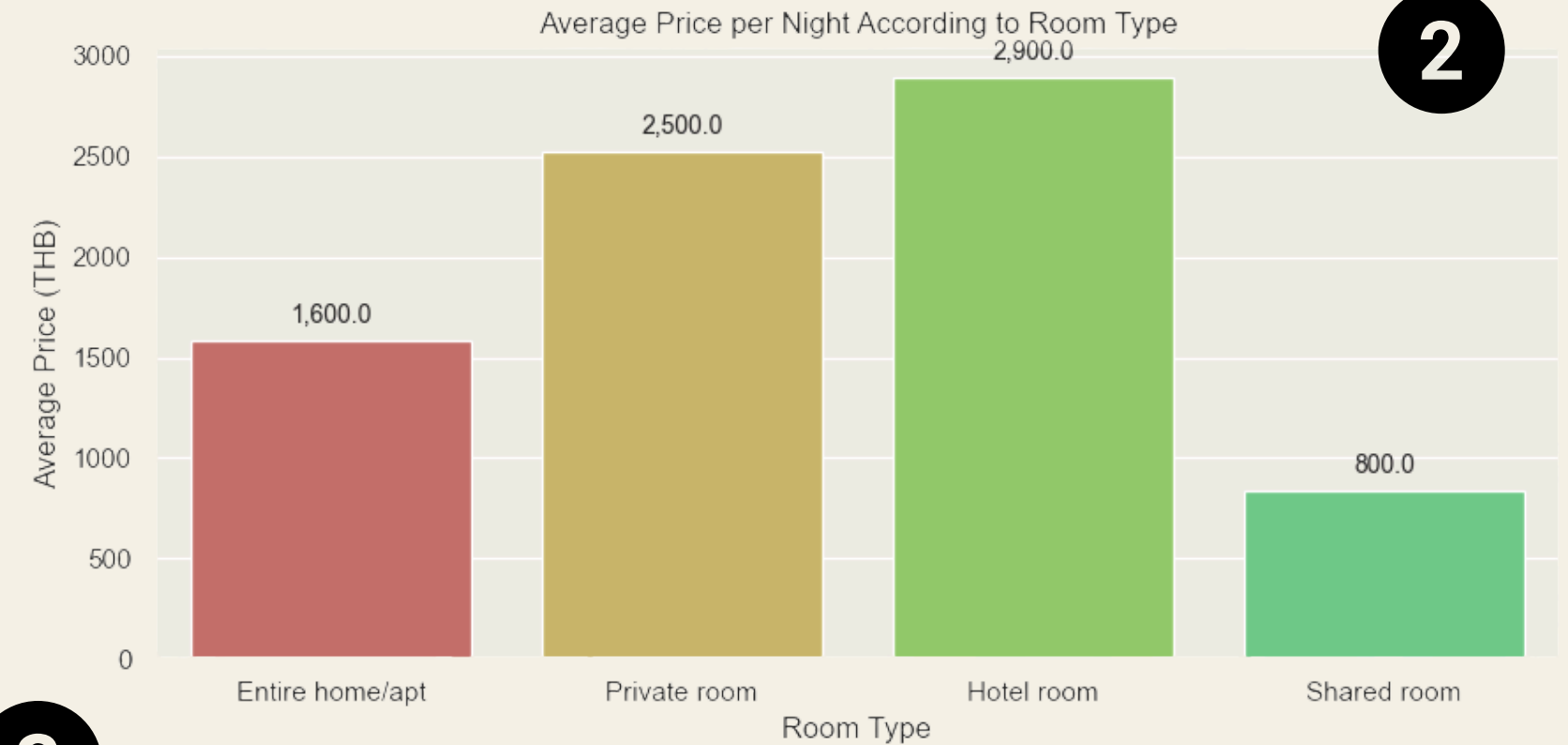
Exploratory Data Analysis

Misaligned Pricing

1



2



3

Unrealistic Price Per Night

- **Very few overpriced listings** – only 37 listings are priced above the high threshold (~25M IDR / 49K THB).
- **Most listings are low-priced** – 3,633 listings sit just above the minimum reasonable price (~70K IDR / 138 THB).

Key Findings:

(Price_per_night)
Huge Outliers (Entire Home/Apt):
1.0 x 1e6 : 1,000,000 THB

(Price_per_night)
Mismatch Average Price per Night
Avg 1,600 for Entire Home/Apt - Cheap
(Facility >< Price)

Unrealistic Price Per Night
(Mismatch or Loss Profit-Making)

Exploratory Data Analysis

Low Engagement

Hosts with more than 5 Listings

1

Host ID	Avg Price	Avg Number of Reviews	Avg Availability (365 Days)
153793	1983.538462	21.538462	124.769231
263049	1554.428571	39.857143	346.571429
323158	21933.500000	7.166667	244.500000
420188	2907.666667	80.222222	62.444444
822284	1302.857143	13.714286	136.428571

2

3158 Hosts

High availability (> 200 days) and Low Bookings Review (< 5 reviews)

Many Listings

Have < 5 reviews illustrate lower search ranking & customer trust

92.13%

Hosts with No Monthly Reviews, <1 review/month

Key Findings:

- **Host (1) 323158:** High price, low reviews → likely overpriced.
- **Host (2) 263049:** Good reviews, high availability → underutilized despite demand.

No Regular Bookings

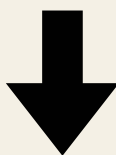
Competition is High

Exploratory Data Analysis

Uneven Host Performance and Low Occupancy Rate

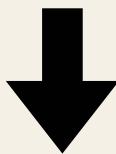
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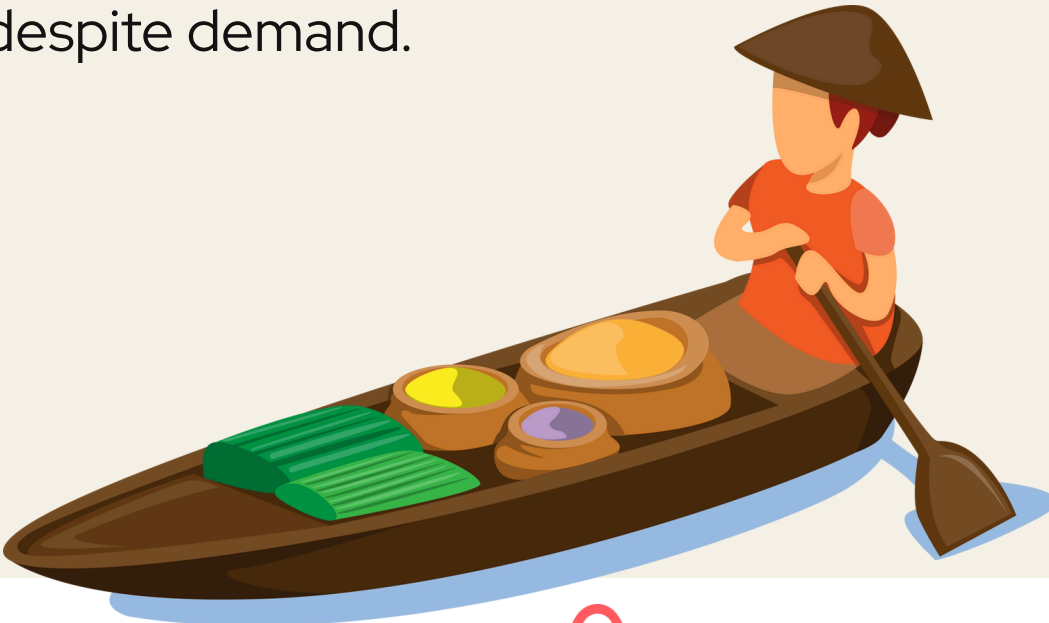


- 1). **Host 420188:** 80+ reviews, low availability → efficient pricing and strong demand.
- 2). **Host 323158:** Avg. price \$21,933 low reviews, high availability → likely overpriced, low occupancy.
- 3). **Host 263049:** Low prices, strong reviews, high availability → underperforming despite demand.

Key Findings:

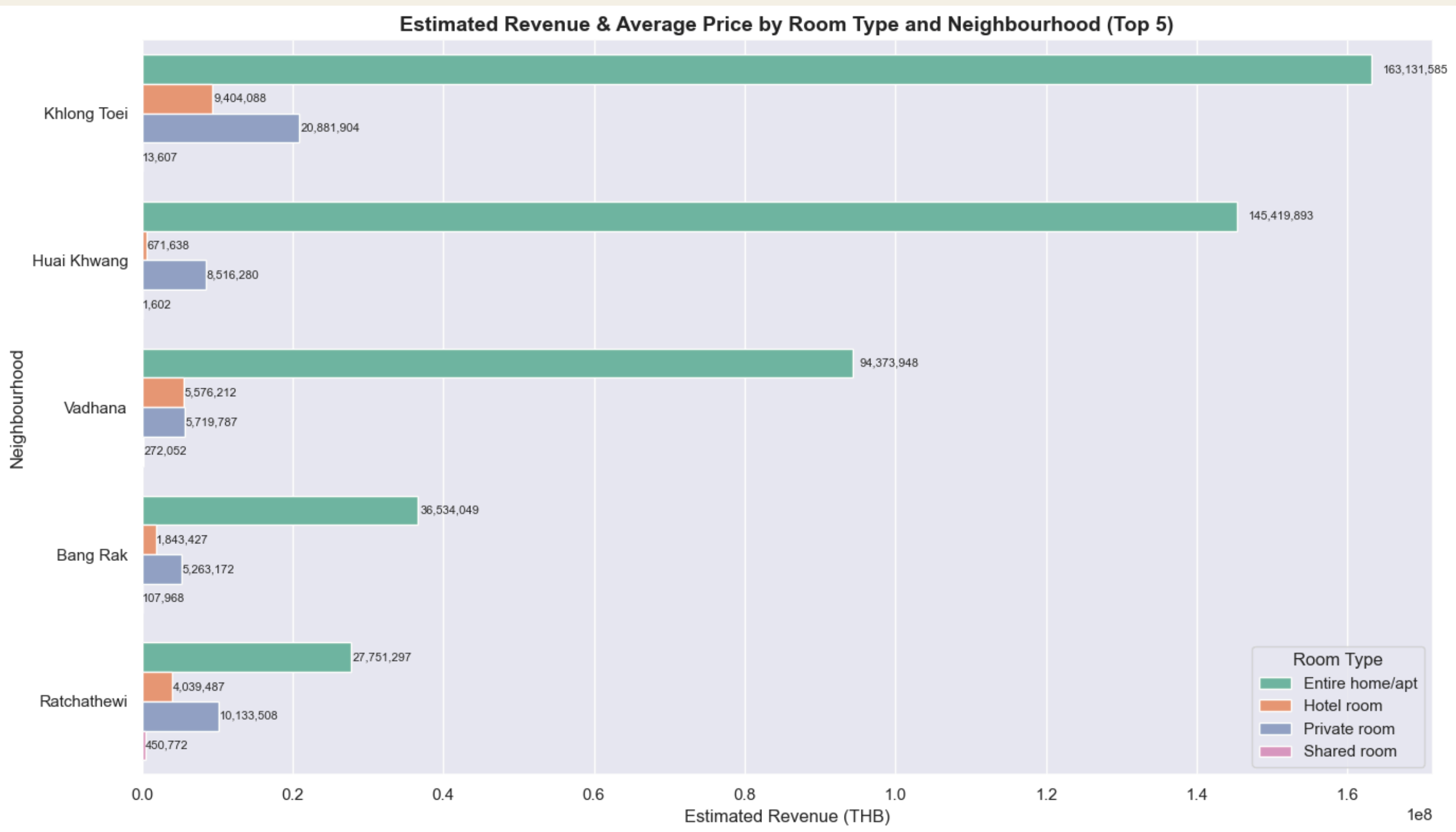


Uneven Host Performance



Exploratory Data Analysis

Top Revenue by Neighbourhood

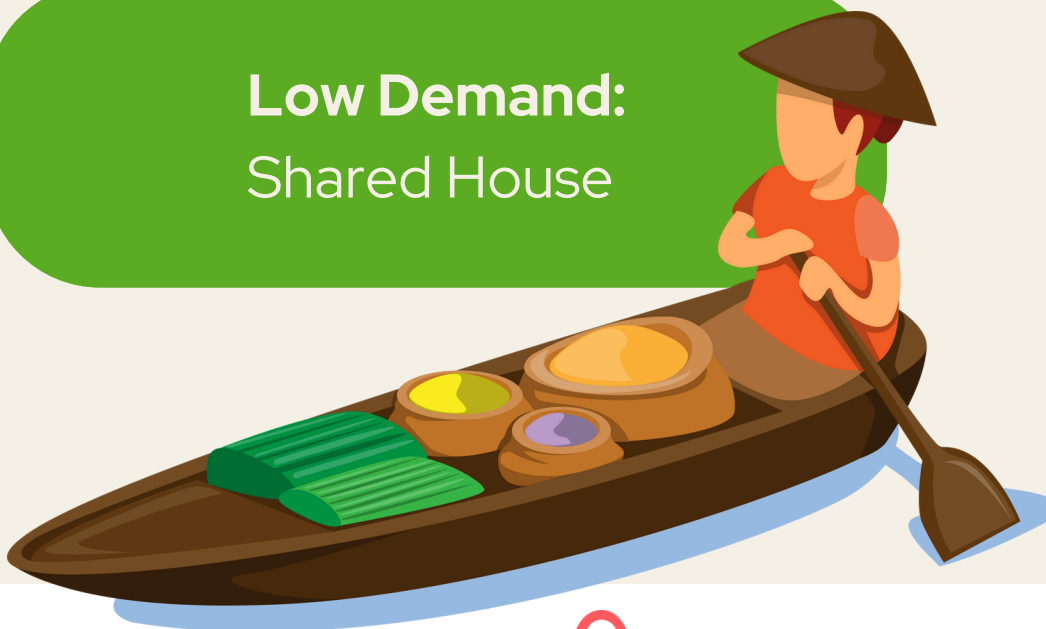


Room Type Domination:
Entire Home/Apt

Location Patterns:
Place near Sukhumvit (City)

Competitive Patterns:
Vadhana District,
Private Room and Hotel Room

Low Demand:
Shared House



ACTIONABLE PLAN RECOMMENDATION

Actionable Plan Recommendation

1). *Dynamic Pricing & Revenue Optimization Program*

Problem addressed: Misaligned Pricing

Data basis: Price Outliers, Unrealistic Price Per Night, Mismatch the Room Type Pricing

Action	Stakeholders	Expected Impact	KPI
Introduce auto-pricing nudges for hosts with >200 idle days	Airbnb Product Team	+10–15% occupancy uplift	Days booked / 365
Deploy “Smart Pricing Benchmark” report per district	Airbnb Analytics + Customer Experience Team	Reduces pricing mismatch across neighbourhoods & room type	Avg. price vs demand curve
Offer revenue-share incentive if hosts adopt dynamic pricing	Airbnb Product Team	Faster adoption rate	% listings using auto-pricing

Actionable Plan Recommendation

2). Host Review & Engagement Acceleration

Problem addressed: Many listings have <5 reviews, even 92.13% with no monthly reviews.

Data basis: Low Engagement EDA, room with long idle days < 5 reviews.

Action	Stakeholders	Expected Impact	KPI
Post-stay “auto review reminder” workflow (Email)	Airbnb CMO and Customer Experience Team	+20–25% review conversion	% of stays leaving reviews
Badge system: “Fast Responding Host”, “Top Reviewed”, “Most Influential Host”	Airbnb Product	Higher conversion in search filters	Search (Booking CTR)
Enable guest incentive: 5% coupon for verified review	Airbnb CMO	More verified review volume	New reviews per month

Actionable Plan Recommendation

3). Portfolio Host Performance Coaching (for hosts with >5 listings) & Low Occupancy Rate

Problem addressed: Uneven performance across listings even under same host ID, Low occupancy rate.

Data basis: Hosts with 5+ properties but huge spread in price / availability / reviews, long idle days.

Action	Stakeholders	Expected Impact	KPI
Monthly performance dashboard sent to hosts	Airbnb Operational Team	Standardizes revenue management per host	Host RevPAR variance ↓
Host Academy: micro-courses on pricing, SEO, photo quality	Airbnb Learning Team	Reduces under-optimized listings	% listings >60% occupancy
Partnership: Thai tourism board, airlines, festivals	Airbnb Product Team	Bundled stay packages	Avg. length of stay ↑

APPENDIX

Data Understanding

Data Cleaning

Before Cleaning:

- 17 columns, 15,854 rows with inconsistent formatting.
- Missing values in name(8), host_name(1), last_review(5790)
- Duplicates found across identical coordinates.
- Zero or invalid prices.

Cleaning Steps:

- Standardized column names → consistent schema.
- Filled missing host names (Unknown), reviews (0), last review (NaT)
- Analyze records with multiple listings to find the right host name with the same properties (Pakaphol)
- There are 7 listings with no name, for further consideration (6 Private Room, 1 Entire Home/Apt)
- Removed listings with price = 0.
- Dropped duplicates based on location and review date.
- Converted price to numeric and ensured correct types.
- Add new columns (last_review(Year)), (price_per_night), and (estimated_review) for the further analysis.

After Cleaning:

- Dataset reduced to reliable, unique listings.
- Null values < 0.5%.
- Ready for accurate analysis.

Data Transformation

Data	Transformation Formula	Data Transformation
price & number_of_reviews	$\text{estimated_revenue} = \text{price} * \text{number_of_reviews}$	Estimated Revenue
availability_365	$\text{occupancy_rate} = 1 - (\text{availability_365} / 365)$	Occupancy Rate
last_review	extract year / month in Tableau (DATENAME('month',[last_review]))	lastReviewMonth, lastReviewYear
room_type	No transform (dimension)	roomType
neighbourhood	No transform (dimension)	neighbourhood
reviews_per_month and host_id	reviews_per_month < 1 with nunique of host_id	low_review_hosts
price	Create bin (Low / Medium / High) in Tableau	Price Level
id & host_id	used to create distinct counts in Tableau	Unique Hosts and Listings
reviews_per_month and room_type	$\text{SUM}([\text{reviews_per_month}]) / \text{COUNT}([\text{Room Type}]$	Monthly Reviews Rate